

Problem Statement 1: Doctor Shift Assignment using Backtracking

Assign three doctors (D1, D2, and D3) to three hospital shifts (Morning, Afternoon, and Night) such that:

D1 cannot work the Night shift.

D2 must be assigned to a shift before D3 (Morning < Afternoon < Night).

Only one doctor can be assigned to each shift.

D3 cannot work the Morning shift.

Every doctor must be assigned exactly one shift.

Apply the Backtracking Search Algorithm to find a valid assignment. Show all intermediate steps, constraint checks, and state the final valid schedule.

Problem Statement 2: Robot Path Finding using BFS Search

A robot operates in a 5×5 grid environment and must move from the Start (S) position to the Goal (G) position. Some cells contain obstacles that cannot be crossed.

Constraints:

The robot can move only Up, Down, Left, or Right.

Each move has a cost of 1.

The robot cannot pass through obstacles.

Use the Manhattan Distance heuristic.

Apply the Breadth-First Search (BFS) algorithm to show the step-by-step node expansion, queue updates, and determine the optimal path and total cost.

Problem Statement 3: Autonomous Rescue Robot using AI Search

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The building is represented as a grid of rooms. The robot starts at position S and must reach the survivor at position G in a partially observable environment.

Constraints:

Some cells are blocked by obstacles.

The robot can observe only adjacent cells.

Each movement has a cost of 1, and entering a risky cell adds an extra cost of 2.

The robot must minimize the total cost while navigating safely.

The robot updates its knowledge dynamically during navigation.

Analyze the constraints, develop a solution using an appropriate search strategy (such as Uniform Cost Search or an Online Search Agent), explain the AI concepts involved, and justify the chosen approach.